DualSight-Fire: A Cognitive Dual-Decoder Network for Robust Fire Segmentation in Multispectral Remote Sensing Images

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***Abstract*—Small fire segmentation in remote sensing imagery is a critical yet challenging task due to the extreme foreground–background imbalance and the interference from complex background clutter. To this end, we propose a novel dual-decoder network, DualSight-Fire, to accurately detect small fire spots. The proposed method consists of a segmentation decoder and a reconstruction decoder. The reconstruction branch incorporates a contextual background modeling block (CBMB) to explicitly model background features and capture long-range dependencies. To enhance the discriminability of fire targets, a feature discrepancy fusion module is proposed to suppress background noise by exploiting the discrepancy between the segmentation and reconstruction features. Moreover, we propose a spatial-focal loss (SFL), which dynamically weights flame regions based on their area and shape, thereby achieving targeted learning optimization for tiny targets and boundary information. Extensive experiments on the Active Fire and Land8Fire datasets demonstrate that the proposed method outperforms state-of-the-art approaches, obtaining the highest $F1$-score and IoU. The code is available at https://github.com/HIAS-Zhao/fire-segmentation**

***Index Terms*—Background reconstruction, dual-decoder network, fire segmentation, multispectral infrared image, spatial-aware loss.**

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I. Introduction

WILDFIRE detection using remote sensing imagery is a critical task for environmental monitoring and disaster early warning [1], [2], [3]. Satellite-based sensors, particularly in thermal infrared and short-wave infrared bands, enable the identification of active fire spots as localized thermal anomalies against the Earth's surface. However, achieving reliable automated detection is severely hindered by a high rate of false alarms. Numerous nonfire objects, such as industrial heat sources, urban lighting, sun glint on water bodies, and hot bare soil, can exhibit similar radiometric signatures to fires, appearing as ambiguous high-radiance targets in imagery [4]. Distinguishing true fires from these prevalent confounders remains a core challenge.

Early efforts to address this challenge relied on physically motivated thresholding and contextual active fire detection algorithms. Representative examples include the MODIS contextual fire detection algorithm [5] and the VIIRS 375 m active fire product [4], which exploit brightness temperature contrasts between candidate pixels and their local neighborhoods. Although effective for large and hot fires, such methods rely on hand-crafted thresholds and stationary background assumptions, leading to low sensitivity to small fires with weak thermal signatures and elevated false alarms over heterogeneous surfaces. With the rise of deep learning, fully convolutional architectures [6], [7], [8] have been adapted to fire segmentation [9], [10], [11], [12], [13], replacing handcrafted features with end-to-end representation learning. Yet, the limited receptive field of convolutions constrains their ability to model the long-range semantic context that is required to disambiguate fire pixels from spectrally similar confounders such as industrial heat sources or hot bare soil. Transformer-based models [14], [15], [16], [17], [18] alleviate this issue by capturing global dependencies through self-attention, and have shown promising results on remote sensing segmentation tasks. Nevertheless, all of the above paradigms share a common formulation: a single-stream encoder–decoder produces pixel-wise classifications without explicitly modeling what the nonfire background should look like. Under the extreme foreground–background imbalance [19] and severe semantic confusion characteristic of small fire detection, such single-stream networks lack an explicit

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background-aware mechanism to assess each candidate anomaly in relation to the surrounding scene semantics, and therefore remain prone to misclassifying look-alike thermal artifacts as fires.

This limitation motivates a background-aware perspective inspired by how human analysts distinguish true fires from visually similar nonfire anomalies. As shown in Fig. 1, we investigated the human cognitive process involved in making such complex judgments. Individuals do not conclude the presence of fire based solely on the detection of a bright pixel. Instead, the process involves a sequential two-stage reasoning mechanism. First, the analyst's attention is captured by a local anomaly, such as a potential foreground candidate. This initial spotting entails an adaptive mechanism: analysts instinctively increase their sensitivity to tiny, inconspicuous fire spots to avoid missed detections, while simultaneously sharpening their focus on the ambiguous boundaries of these anomalies to ensure precise delineation. Subsequently, a critical background confirmation stage occurs. Here, the analyst interprets the broader context not merely as spatial surroundings, but as semantic scene content. They assess whether the anomaly is situated within a plausible wildfire context or coincides with areas typically associated with anthropogenic heat sources. This conscious cross-referencing of the candidate against the underlying land cover or scene type is fundamental for accurate diagnosis [20].

Conceptually, science and neuroscience research supports this staged visual reasoning. The human visual system is adept at rapid scene gist extraction, categorizing a scene within milliseconds, which in turn primes expectations for likely objects within it [21]. This top–down influence of scene semantics guides attention and aids in resolving perceptual ambiguities [22]. This biological evidence underscores that reliable detection is not a product of isolated feature analysis, but of an integrative process where candidate identification is verified against a semantically understood background.

Inspired by this human cognitive process, we propose a novel framework, DualSight-Fire, which explicitly emulates the dual-stage "foreground-spotting and background-confirmation" mechanism. Unlike conventional segmentation networks that perform pixel-wise classification within a single stream, our architecture features two dedicated decoding branches with distinct and complementary functions. The segmentation decoder serves as the "foreground spotter." Its primary task is to sensitively identify all potential bright thermal anomalies from the input image. In parallel, the reconstruction decoder functions as the "background confirmer." Its objective is not to simply reproduce the input image, but to learn and reconstruct a clean representation of the background scene—that is, the underlying landscape devoid of fire anomalies. This branch aims to encapsulate the intrinsic structure and semantic content of typical nonfire terrains.

The core innovation of our framework lies in the feature discrepancy fusion module (FDFM), which implements the confirmation logic. This module strategically compares the feature maps from the foreground-spotting branch with the semantic context from the background-reconstruction branch. Through this comparison, features corresponding to true wildfires are enhanced, while features arising from common confusers are suppressed. This process directly mimics the analyst's cognitive act of cross-referencing a candidate anomaly against its broader scene context.

Furthermore, to incorporate the adaptive attention control exhibited by human experts into the model optimization, we introduce a spatial-focal Loss (SFL). This loss function integrates spatial awareness by dynamically assigning higher penalization weights to small and irregularly shaped fire instances. This ensures the model maintains high sensitivity toward these challenging targets and their ambiguous boundaries, analogous to the focused scrutiny of a skilled analyst.

To validate the proposed method, we conduct comprehensive experiments on two Landsat-8 multispectral infrared datasets: Active Fire [9] and Land8Fire. Under ten random-seed runs on the same data split for each benchmark, DualSight-Fire achieves state-of-the-art mean performance, obtaining $F1$-scores of 95.78% and 94.55% and IoUs of 91.89% and 89.67% on Active Fire and Land8Fire, respectively. Paired statistical significance tests confirm the reliability of these improvements. Ablation studies further demonstrate the effectiveness of each key component, particularly the dual-decoder architecture and feature discrepancy fusion module. The proposed SFL also proves superior to conventional loss functions, especially for detecting small fire targets.

The main contributions of this work are summarized as follows.

1) We propose DualSight-Fire, a cognitively inspired network that separates and synergizes foreground candidate spotting with background semantic confirmation for robust fire segmentation, effectively reducing false alarms.
2) We design a feature discrepancy fusion module that implements confirmation logic by contrasting multi-branch features, suppressing responses inconsistent with fire semantics.
3) We introduce an SFL to address extreme class imbalance by focusing on small and boundary fire pixels.
4) Extensive qualitative and quantitative experiments prove the superiority of our approach.

II. Related Work

A. Fire Segmentation in Remote Sensing

Early fire segmentation methods primarily relied on spectral indices and physical thresholding. The MODIS-era contextual fire algorithms proposed by Kaufman et al. [23] and Giglio et al. [5], [24], together with the VIIRS 375 m active-fire product [4], utilize brightness temperature differences between fire spots and the background to identify anomalies. To address the limitation of static thresholds under heterogeneous backgrounds, Koltunov et al. [25] further introduced

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a nonlinear multitemporal prediction scheme for thermal imagery. While effective for large-scale fires, these classical methods still struggle with small fires with weak thermal signatures and suffer from high false-alarm rates in complex environments.

With the rise of deep learning, convolutional neural networks have become the dominant paradigm. Building upon generic semantic segmentation backbones such as FCN [6], U-Net [7], and DeepLabv3+ [8], researchers adapted these architectures to perform fire segmentation. de Almeida Pereira et al. [9] fine-tuned U-Net on Landsat-8 image, demonstrating superior performance over traditional thresholding methods.

Then, Harkat et al. [10] evaluated DeepLabv3+ with different encoders for fire images. Yar et al. [26] introduced an efficient CNN with modified attention mechanisms. Wu et al. [27] proposed a fast stochastic configuration network for flame recognition. Hu et al. [28] utilized deep semantic segmentation models for large-scale burn severity mapping. Bo et al. [11] proposed BASNet for the segmentation of the burned area, focusing on boundary awareness.

Recent studies utilized transformer-based models. Following the success of generic transformer segmentation frameworks such as Swin Transformer [14] and SegFormer [15], and their remote-sensing extensions [16], Zhao et al. [17] applied transformer networks to time-series satellite images for active fire detection, and Agarwal et al. [18] used vision transformers for improved wildfire detection and characterization. Despite these advances, handling the extreme class imbalance remains a challenge. Specialized networks like FPS-U^{2}Net [12] and FDE U-Net [13] proposed to enhance feature aggregation and flame details, yet they may still struggle with complex semantic confusion.

B. Small Object Segmentation

Semantic segmentation has advanced rapidly, yet segmenting small objects remains a significant challenge, primarily due to extreme class imbalance and the loss of spatial detail at low resolutions. To preserve small-target features, classical architectures such as the feature pyramid network (FPN) [29] aggregate multiscale features in a top–down manner, while HRNet [30] maintains high-resolution representations throughout the network and has become a strong baseline for small-object dense prediction. On the loss side, focal loss [31] was introduced to down-weight easy negatives and focus training on hard examples. However, standard focal loss treats all pixels equally and fails to account for the spatial characteristics of targets, which is crucial for small object segmentation. To this end, Luo et al. [19] proposed a spatial-temporal-aware network for infrared small target detection, leveraging multiframe information. Our proposed SFL advances this direction by directly integrating area and shape constraints into the weighting mechanism.

C. Dual-Stream Networks for Background Modeling

The idea of decoupling foreground and background cues through parallel branches can be traced back to the two-stream convolutional networks of Simonyan and Zisserman [32], and was later extended to explicit background semantic modeling by the context encoding network of Zhang et al. [33]. Building on this line, dual-stream or multitask networks have been successfully applied to scenes requiring background suppression, such as anomaly detection and shadow removal. Lu et al. [20] proposed a dual-branch network for cloud and cloud shadow segmentation. Wang et al. [34] developed a dual-branch model with adaptive feature fusion for UAV small target detection. Recent studies have further demonstrated the effectiveness of biological-inspired or multibranch designs in complex tasks: Xu et al. [35] employed an asymmetric dual-branch network for real-time segmentation; Wang et al. [36] utilized joint learning for underwater object detection; and Zeng et al. [37] proposed a dual-decoding branch structure for image manipulation localization. Hao et al. [38] utilized background suppression with proximal gradients for infrared small target detection. Our method separates the modeling of the foreground and background by introducing a dedicated reconstruction branch, thereby reducing ambiguity.

In summary, classical thresholding and contextual fire detection algorithms [4], [23], [24], [25] laid the physical foundation for fire monitoring, but they often struggle with small fires with weak thermal signatures under heterogeneous backgrounds; CNN and transformer-based segmentation networks [9], [10], [17], [18] substantially improve foreground learning, yet still treat the background as an implicit negative class and offer limited resistance to fire-like distractors; small-object-oriented designs [19], [29], [30], [31] alleviate the class-imbalance issue but do not explicitly model what a nonfire background should look like; existing dual-stream architectures [20], [32], [33], [34] mainly target generic background suppression rather than verifying fire candidates against a learned background prior. This leaves a clear gap: a unified framework that simultaneously preserves high sensitivity to tiny fire candidates and performs explicit, learnable background confirmation. The proposed DualSight-Fire, together with its feature discrepancy fusion module and SFL, is designed to fill this gap.

III. Method

A. Architecture

To effectively address the challenge of small fire spot segmentation in complex remote sensing imagery, we design a dual-decoder network that simultaneously performs fire segmentation and background reconstruction. The overall architecture is depicted in Fig. 2.

As shown in Fig. 2, DualSight-Fire is organized into four sequential steps. Step1 extracts multiscale semantic features from the input image through a shared encoder. The encoded features are then dispatched to two parallel decoding branches: Step2-1 progressively upsamples the features to produce the fire probability response **S**, while Step2-2 first refines the encoded bottleneck features through an independent Swin-transformer block (STB) and then feeds them into four cascaded contextual background modeling blocks (CBMBs) to yield the background response **R**. Finally, Step3 fuses

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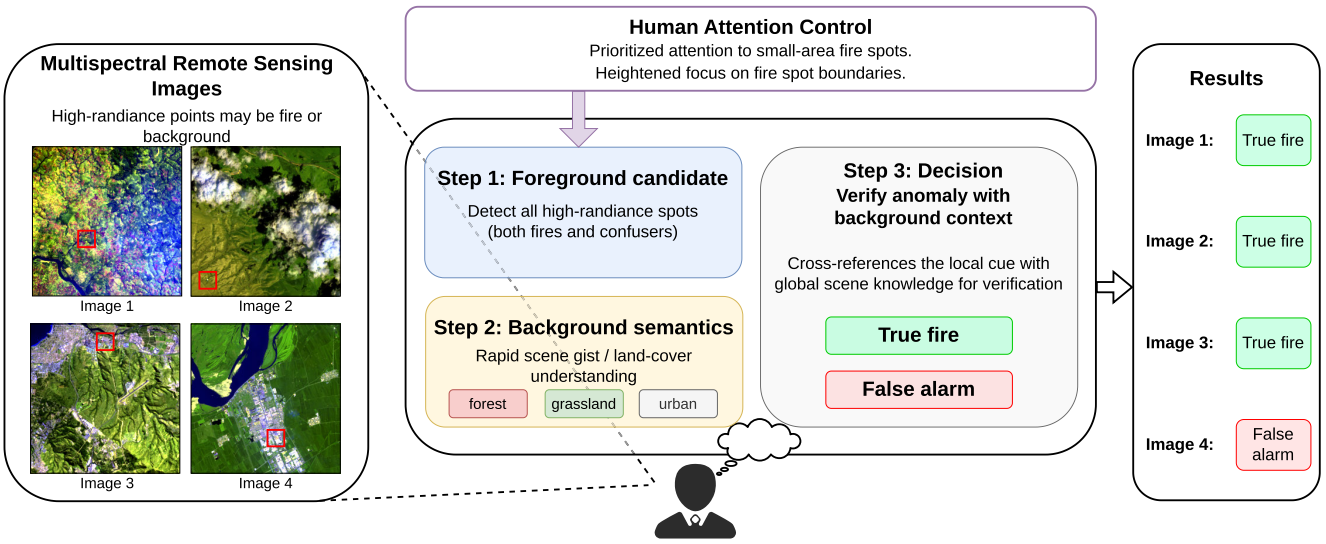


Fig. 1. Illustration of the human cognitive strategy for fire segmentation. The process emulates a dual-stage reasoning mechanism: the rapid identification of salient anomalies (foreground spotting) followed by their verification against the semantic scene context (background confirmation). This approach robustly distinguishes true fires from look-alike confusers.

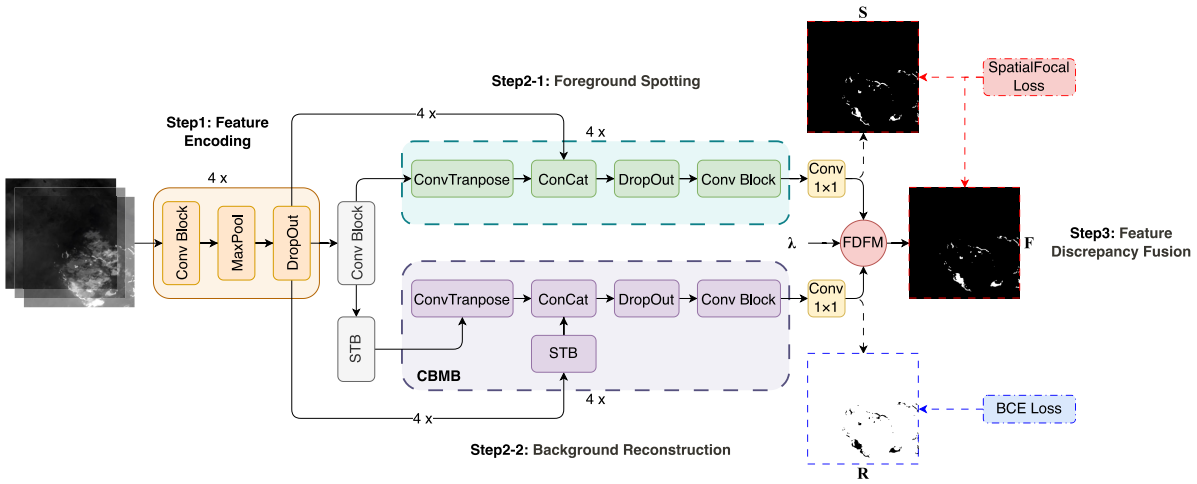


Fig. 2. Overview of the DualSight-Fire method.

S and **R** through the feature discrepancy fusion module (FDFM) to suppress fire-like distractors and produce the final prediction **F**.

1) Feature Encoding and Foreground Spotting: The shared encoder follows a standard CNN backbone consisting of four cascaded stages, each composed of a Conv block, a MaxPool layer, and a DropOut layer, producing multiscale features for the two decoding branches. The internal structure of the Conv block is illustrated in Fig. 3(b).

The segmentation decoder adopts a standard U-Net decoder structure, comprising four consecutive up blocks. Each up block performs feature upsampling and concatenation with the corresponding encoder features via skip connections, followed by convolutional layers to recover spatial details and generate the foreground features **S**.

2) Background Reconstruction: Parallel to Step2-1, the reconstruction decoder explicitly models background semantics and outputs the background response **R**. Before entering the decoder, the bottleneck features from the encoder are first refined by an independent STB to inject global context. The refined features are then progressively upsampled by four cascaded CBMBs with an identical structure, as illustrated in Fig. 2.

Each CBMB consists of an STB followed by an up block. The STB refines the skip-connected features from the corresponding encoder stage using shifted window-based multihead self-attention, providing a robust global context that complements the local features learned by CNNs. The STB-enhanced skip features are then concatenated with the upsampled features from the previous stage and processed by the up block to recover spatial resolution. The internal structure of the STB is illustrated in Fig. 3(a).

3) Feature Discrepancy Fusion: To refine the segmentation results, we propose the feature discrepancy fusion module (FDFM). This module exploits the complementary nature of the two branches at the prediction level. By subtracting

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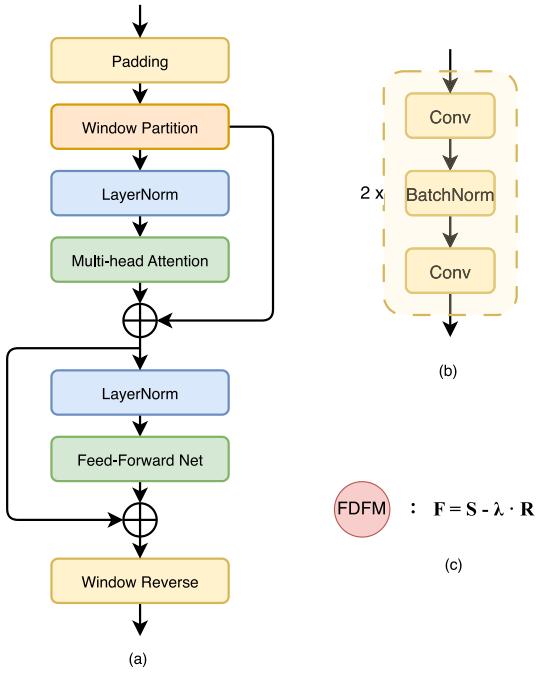


Fig. 3. (a) STB. (b) Conv block, the basic unit of the encoder in Step1. (c) Feature discrepancy fusion module (FDFM).

the weighted background reconstruction features **R** from the segmentation features **S**, we can effectively suppress false positives (FPs) caused by bright objects or background artifacts. The final fused features **F** are obtained as

$$\mathbf{F} = \mathbf{S} - \lambda \cdot \mathbf{R} \tag{1}$$

where λ is a learnable parameter that adaptively controls the contribution of the background suppression.

Conceptually, **R** acts as a soft, per-pixel "background-likeness" map produced by the reconstruction decoder, which is supervised to recover the nonfire semantics of the scene such as urban built-up areas, factory roofs, road networks, terrain textures, and high-albedo bare ground. The subtraction in (1) can therefore be read as a learnable per-pixel semantic veto: at locations where the foreground response **S** is high and the reconstruction **R** is also high, the candidate features are well explained by a known background category and the fused response **F** is sharply attenuated, so fire-like confusers including industrial and factory heat sources, urban built-up areas, hot bare soil, and sun glint on water are systematically suppressed; conversely, true fires that do not match any learned background mode keep **R** small and therefore survive the subtraction. The FDFM thus behaves as an explicit, learnable false-alarm filter whose subtraction strength is adaptively controlled by λ.

B. Spatial-Focal Loss

Remote sensing fire segmentation severely suffers from a significant foreground–background class imbalance. During training, a large number of small fire pixels are easily overwhelmed by the extensive background regions. Standard losses like binary cross-entropy (BCE) treat all pixels equally, leading to model bias toward the background. While focal loss addresses class imbalance, its static focusing parameters fail to account for the intraclass variance (e.g., size and shape) of fire targets. To bridge this gap, we propose a spatial-aware loss function that dynamically modulates the learning focus based on the spatial properties of individual fire instances.

1) Area-Aware Weighting Strategy: Small fire spots are often the most critical yet difficult targets to detect. In standard training, the gradient contribution from a few small fire pixels is negligible compared to large fire zones or vast background areas. To rectify this, we introduce an area-aware weighting mechanism that inversely correlates the loss weight with the target size.

Specifically, we perform connected component analysis (CCA) on the ground truth masks to identify individual fire instances. Let $\mathcal{C} = \{C_1, C_2, \ldots, C_K\}$ denote the set of K connected fire components in the current image (to preserve local contrast), and A_k be the pixel area of component C_k. We define a dynamic balancing factor α_j for each pixel j as follows:

$$\alpha_j = \begin{cases} \dfrac{A_{\max}}{A_{\max} + A_{\mathcal{T}(j)}}, & \text{if } y_j = 1 \\ \alpha_{\text{bg}}, & \text{if } y_j = 0 \end{cases} \tag{2}$$

where $\mathcal{T}(j)$ maps a foreground pixel j to its corresponding component index, $A_{\max} = \max_k A_k$ is the area of the largest fire instance in the image, and α_{bg} is the background weight. α_{bg} was empirically set to 0.5, and a sensitivity analysis was conducted. This hyperbolic-like scaling ensures that pixels belonging to tiny fire spots ($A_{\mathcal{T}(j)} \ll A_{\max}$) yield weights approaching 1.0, while the dominance of large fire areas is mitigated as their weights decay toward 0.5. For images containing no fire spots ($K = 0$), we default $\alpha_j = \alpha_{\text{bg}}$ for all pixels.

2) Shape-Aware Focusing Mechanism: Precise boundary delineation is essential for fire monitoring. Pixels near the fire boundaries are often ambiguous and harder to classify than those at the fire center. To enhance the model's sensitivity to edge details, we propose a shape-aware focusing mechanism.

Concretely, for every fire component C_i, we compute the Euclidean distance of each internal pixel to the nearest background pixel. To ensure scale invariance across different fire sizes, we normalize these distances. Let $D(j)$ be the raw distance for pixel $j \in C_i$, and $D_{\max,i}$ be the maximum distance within component C_i. The normalized distance is $d_j = D(j)/D_{\max,i}$, where $d_j \in [0, 1]$.

We then dynamically adjust the focusing parameter γ of focal loss. The shape-aware γ_j is defined as

$$\gamma_j = \begin{cases} \gamma_0 \cdot (1 - d_j) + 1, & \text{if } y_j = 1 \\ 1, & \text{if } y_j = 0 \end{cases} \tag{3}$$

where γ_j is set to 1 for images that contain only background ($y_j = 0$). In cases where fire points are present, γ_0 serves as a hyperparameter controlling sensitivity; boundary pixels ($d_j \to 0$) are assigned higher γ values, increasing their contribution to the loss. The center pixels ($d_j \to 1$) have lower γ values, reducing their relative importance. This gradient-like focusing mechanism encourages the model to refine the

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TABLE I
COMPARISON OF FIRE PIXEL DISTRIBUTION ACROSS DATASETS

Dataset	Fire pixel count range (Number of images/Percentage)					Total
	0	Level I (1–10)	Level II (11–100)	Level III (101–1000)	Level IV (>1000)	
ActiveFire Dataset [9]						
Training	0 (0.00%)	28,665 (85.51%)	4,433 (13.22%)	422 (1.26%)	3 (0.01%)	33,523
Validation	0 (0.00%)	7,109 (84.83%)	1,146 (13.68%)	123 (1.47%)	2 (0.02%)	8,380
Test	0 (0.00%)	35,740 (85.29%)	5,612 (13.39%)	540 (1.29%)	10 (0.02%)	41,902
Land8Fire Dataset [39]						
Training	8,835 (95.66%)	200 (2.17%)	112 (1.21%)	74 (0.80%)	15 (0.16%)	9,236
Validation	2,213 (95.84%)	43 (1.86%)	29 (1.26%)	22 (0.95%)	2 (0.09%)	2,309
Test	10,965 (94.96%)	266 (2.30%)	187 (1.62%)	107 (0.93%)	22 (0.19%)	11,547

segmentation boundaries. γ_0 was empirically set to 2, and a sensitivity analysis was conducted.

3) Formulation: Combining the area-aware weight α_j and the shape-aware focusing parameter γ_j, the total SFL is formulated as the average loss over all N pixels in the image

$$\mathcal{L}_{\text{Spatial-focal}} = -\frac{1}{N}\sum_{j=1}^{N}\alpha_j\left(1-p_{t,j}\right)^{\gamma_j}\log\left(p_{t,j}\right) \tag{4}$$

where $p_{t,j}$ is the predicted probability of the true class for pixel j.

C. Joint Optimization Objective

To train the DualSight-Fire network end-to-end, we construct a multitask objective function that synergizes the distinct goals of fire detection and background modeling. The total loss $\mathcal{L}_{total}$ is defined as a weighted sum of the losses from the segmentation branch ($\mathcal{L}_{seg}$), the final fusion output ($\mathcal{L}_{fused}$), and the reconstruction branch ($\mathcal{L}_{rec}$)

$$\begin{aligned}\mathcal{L}_{total} = {} & \lambda_{fused}\cdot\mathcal{L}_{\text{Spatial-focal}}(\mathbf{F},Y)+\lambda_{seg}\cdot\mathcal{L}_{\text{Spatial-focal}}(\mathbf{S},Y)\\ & +\lambda_{rec}\cdot\mathcal{L}_{\text{BCE}}\left(\mathbf{R},\bar{Y}\right)\end{aligned} \tag{5}$$

where Y denotes the ground truth fire mask, $\bar{Y} = 1 - Y$ is the background mask, and λ represents the balancing weights. We adopt distinct loss strategies tailored to the inherent characteristics of each branch's task. For fire segmentation ($\mathcal{L}_{fused}, \mathcal{L}_{seg}$), the primary objective is to identify tiny, sparse fire spots plagued by extreme class imbalance and boundary ambiguity. Therefore, we employ the proposed SFL, whose area-aware and shape-aware mechanisms force gradient updates to focus on "hard" examples (tiny spots and boundaries), preventing the model from being overwhelmed by easy negatives. Conversely, the reconstruction branch ($\mathcal{L}_{rec}$) aims to model the global semantic context of the dominant background class. For this dense prediction task, we utilize the standard BCE Loss, as it provides stable gradients for learning global spatial distributions. Applying a focal-style loss here would be counterproductive, as it might excessively penalize common background patterns and hinder robust scene representation.

IV. EXPERIMENTS

A. Datasets

In this study, we utilize the Landsat-8 satellite [40] as the data source. The operational land imager and thermal infrared sensor onboard Landsat-8 provide rich multiband observational data, particularly the short-wave infrared and thermal infrared bands, which exhibit significant spectral responses to high-temperature fire spots. To comprehensively evaluate the performance of the model under different fire scales and complex background conditions, we employed Active Fire [9] and Land8Fire [39] as the evaluation dataset.

As illustrated in Table I, these two datasets demonstrate significant complementarity in their fire spot distribution characteristics. The Active Fire Dataset [9] features a high density of small fire targets, with over 98% of its samples containing fewer than 100 fire pixels and a complete absence of negative samples. This characteristic makes it well-suited for evaluating detection capability for tiny fire spots. In contrast, the Land8Fire Dataset [39] is dominated by background images, where over 95% of samples contain no fire spots. Nevertheless, among the positive samples, those with fewer than 100 fire pixels still account for a considerable proportion. Furthermore, this dataset introduces a substantial number of negative samples, which effectively assesses the ability of the model to suppress false alarms from the background in practical application scenarios.

In terms of spatiotemporal coverage, Active Fire is mainly derived from global Landsat-8 observations acquired in August and September 2020, covering multiple continents except Antarctica and representative fire-prone regions such as the Amazon, Pantanal, Africa, Australia, Siberia, the Parana Delta, and the western United States. Land8Fire is also constructed from Landsat-8 imagery and includes diverse fire and non-fire scenes, providing complementary samples for evaluating robustness under complex land-cover backgrounds, including desert surfaces, urban heat sources, and high-albedo areas.

It should be clarified that both datasets used in this work are based on Landsat-8 imagery and therefore provide discrete observation snapshots rather than continuous temporal sequences. Due to the 16-day revisit cycle of Landsat-8, these data cannot support direct tracking of rapidly evolving fire fronts at the per-frame level. Within the scope of this article, DualSight-Fire is therefore positioned as a per-frame static fire mapping framework, targeting high-precision fire-line delineation at a given observation epoch, burned-area estimation, and postevent analysis, rather than real-time dynamic fire-front monitoring. Nevertheless, even within a single Landsat-8 frame, small fire pixels remain challenging because of their

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TABLE II

MULTISEED COMPARISON OF SEGMENTATION PERFORMANCE ON THE ACTIVE FIRE DATASET. THE PAIRED TEST COLUMNS REPORT TWO-SIDED WILCOXON SIGNED-RANK TEST p-VALUES COMPARING DUALSIGHT-FIRE WITH EACH BASELINE USING SEED-WISE PAIRED $F1$-SCORE AND IOU VALUES

Methods	Precision (%)	Recall (%)	F1-Score (%)	IoU (%)	p for F1	p for IoU
Attention U-Net [41]	93.84 ± 0.82	93.69 ± 0.99	93.76 ± 0.14	88.25 ± 0.25	0.002	0.002
U-Net [7]	93.42 ± 0.53	94.01 ± 0.65	93.71 ± 0.12	88.16 ± 0.22	0.002	0.002
FDE U-Net [13]	94.37 ± 0.63	94.41 ± 0.92	94.39 ± 0.53	89.38 ± 0.95	0.002	0.002
FPS-U^{2}Net [12]	90.05 ± 0.59	92.92 ± 0.52	91.46 ± 0.14	84.27 ± 0.24	0.002	0.002
U-Net++ [42]	94.25 ± 0.65	93.07 ± 0.63	93.65 ± 0.09	88.06 ± 0.16	0.002	0.002
DualSight-Fire (Ours)	$\mathbf{95.74 \pm 0.58}$	$\mathbf{95.82 \pm 0.54}$	$\mathbf{95.78 \pm 0.23}$	$\mathbf{91.89 \pm 0.42}$	–	–

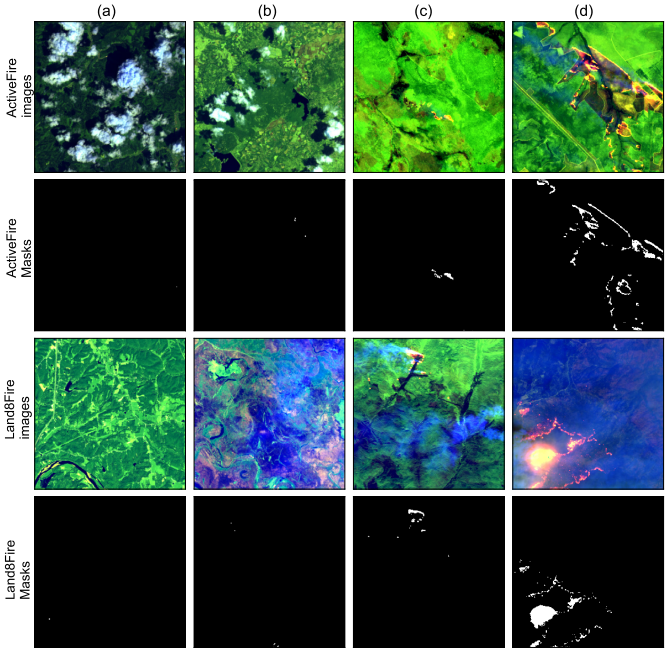


Fig. 4. Example samples from the active fire and Land8Fire datasets organized by fire-pixel count. (a) Level I (1–10 pixels). (b) Level II (11–100 pixels). (c) Level III (101–1000 pixels). (d) Level IV (>1000 pixels).

limited spatial extent, blurred boundaries, and strong background interference; producing reliable static segmentation provides a useful spatial basis for subsequent dynamic analysis when temporally continuous imagery becomes available.

For our experiments, each dataset was randomly split into training, validation, and test sets following a 4:1:5 ratio. Models were trained and tested independently on each dataset to evaluate their performance under specific scenarios (fire-rich versus background-rich). The same split within each benchmark was used for all compared methods and all random-seed runs. Visualization results for some typical samples are provided in Fig. 4.

B. Experimental Setup

All experiments were conducted on the Ubuntu operating system, with hardware consisting of four NVIDIA RTX 5090 GPUs, and the software framework based on PyTorch. For details, please refer to our open-source code.

To ensure fair comparison, the training strategies and parameter settings for all comparison models strictly followed the recommended configurations in the corresponding original publications. The basic settings for attention U-Net [41] and U-Net++ [42] were consistent with U-Net [7], while the training strategy for U-Net followed the settings in the study by de Almeida Pereira et al. [9]. For each method, we conducted ten random-seed runs and report the mean and sample standard deviation of precision, recall, $F1$-score, and IoU. Because the same seed index was used when comparing DualSight-Fire with each baseline, we used the two-sided Wilcoxon signed-rank test as a nonparametric paired test for $F1$-score and IoU. The detailed parameter settings for our DualSight-Fire and other models are presented in the Appendix.

C. Results and Analysis

1) Evaluation on Active Fire Dataset: Table II presents the multiseed quantitative evaluation results of various models on the Active Fire dataset [9]. The proposed DualSight-Fire model achieves the best mean performance across all evaluation metrics, with precision, recall, $F1$-score, and IoU of 95.74%, 95.82%, 95.78%, and 91.89%, respectively. Compared with FDE U-Net [13], the strongest baseline on this benchmark, DualSight-Fire improves the mean $F1$-score from 94.39% to 95.78% and the mean IoU from 89.38% to 91.89%. These results demonstrate that the proposed dual-decoder structure maintains a favorable balance between false-alarm control and fire-pixel recovery under repeated training runs.

Fig. 5 further corroborates the above findings. In the visualization, white pixels denote true positive (TP), black pixels denote true negative (TN), red pixels denote FP, and blue pixels denote false negative (FN). It is observed that DualSight-Fire has the fewest red and blue pixels, indicating the lowest false alarm and missed detection rates.

2) Evaluation on Land8Fire Dataset: To verify the robustness of the model, we conducted experiments on the Land8Fire dataset [39]. As shown in Table III, DualSight-Fire achieves the best mean precision, $F1$-score, and IoU, reaching 96.69%, 94.55%, and 89.67%, respectively. FPS-U^{2}Net obtains a higher mean recall of 95.24%, but its much lower precision leads to substantially lower $F1$-score and IoU. Compared with attention U-Net [41], the strongest Land8Fire baseline in terms of $F1$-score and IoU, DualSight-Fire improves the mean $F1$-score from 93.00% to 94.55% and the mean IoU from

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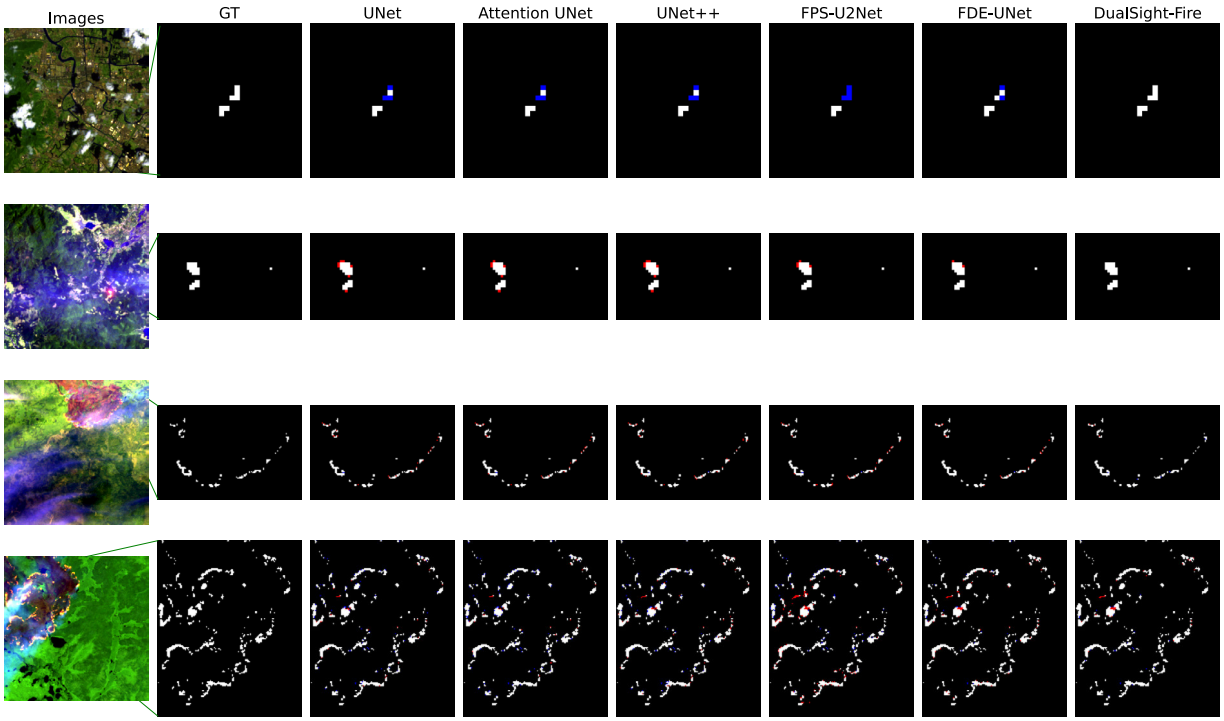


Fig. 5. Comparison of prediction results on the active fire dataset. White pixels represent TP samples, black pixels represent TN samples, red pixels represent FP samples, and blue pixels represent FN samples.

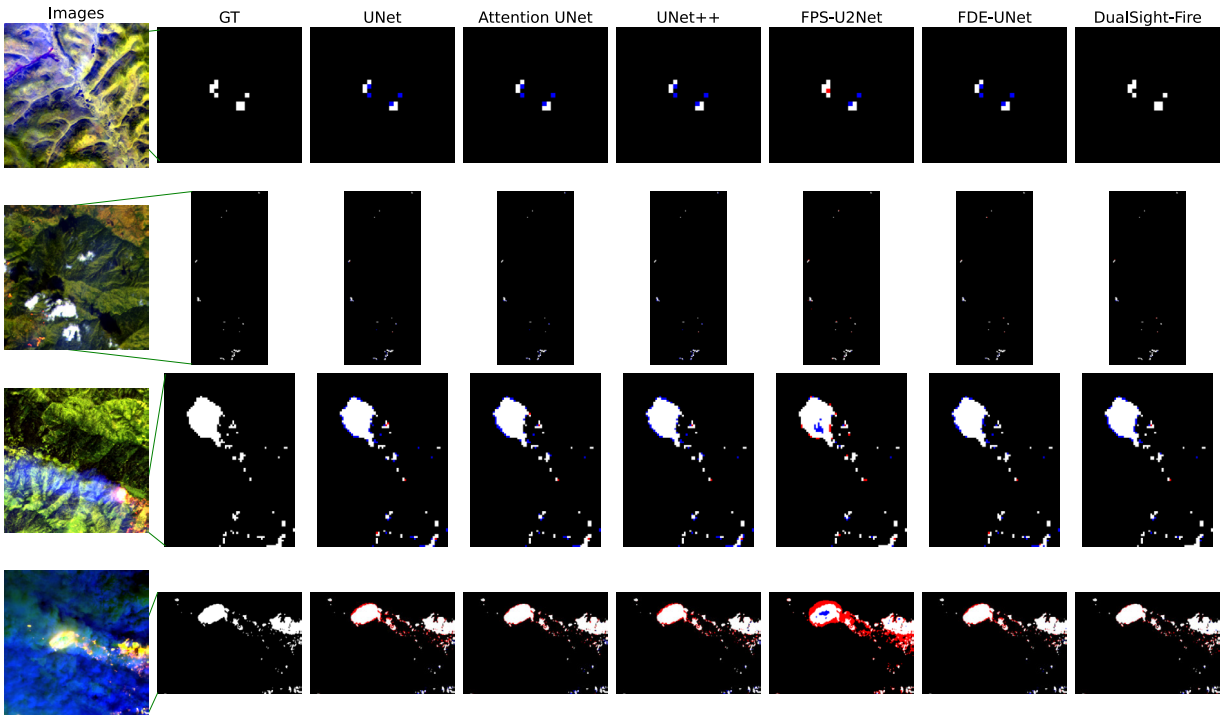


Fig. 6. Comparison of prediction results on the Land8Fire dataset. White pixels represent TP samples, black pixels represent TN samples, red pixels represent FP samples, and blue pixels represent FN samples.

86.92% to 89.67%. This robust performance is mainly attributed to the explicit modeling of background features by the reconstruction decoder, enabling the model to accurately distinguish between bright background interference and real fire spots. Fig. 6 also illustrates the superior performance of the model on the Land8Fire dataset.

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TABLE III

MULTISEED COMPARISON OF SEGMENTATION PERFORMANCE ON THE LAND8FIRE DATASET. THE PAIRED TEST COLUMNS REPORT TWO-SIDED WILCOXON SIGNED-RANK TEST p-VALUES COMPARING DUALSIGHT-FIRE WITH EACH BASELINE USING SEED-WISE PAIRED $F1$-SCORE AND IOU VALUES

Methods	Precision (%)	Recall (%)	F1-Score (%)	IoU (%)	p for F1	p for IoU
Attention U-Net [41]	93.93 ± 1.67	92.13 ± 1.50	93.00 ± 0.73	86.92 ± 1.27	0.002	0.002
U-Net [7]	93.48 ± 2.02	91.22 ± 1.95	92.30 ± 0.46	85.71 ± 0.80	0.004	0.004
FDE U-Net [13]	94.18 ± 1.74	91.52 ± 2.38	92.80 ± 0.90	86.57 ± 1.56	0.010	0.010
FPS-U^{2}Net [12]	72.25 ± 5.46	$\mathbf{95.24 \pm 2.50}$	82.00 ± 2.58	69.56 ± 3.83	0.002	0.002
U-Net++ [42]	93.84 ± 2.60	90.83 ± 1.86	92.27 ± 0.75	85.65 ± 1.29	0.004	0.004
DualSight-Fire (Ours)	$\mathbf{96.69 \pm 0.97}$	92.56 ± 2.53	$\mathbf{94.55 \pm 0.94}$	$\mathbf{89.67 \pm 1.67}$	–	–

TABLE IV

COMPARATIVE RESULTS FOR DIFFERENT FIRE SCALES

Methods	Tiny Fire Group (< 100 pixels)				Large Fire Group (> 100 pixels)			
	F1	IoU	Precision	Recall	F1	IoU	Precision	Recall
U-Net [7]	92.27	85.65	94.38	90.25	89.45	80.91	95.34	84.25
Attention U-Net [41]	92.58	86.18	93.61	91.56	90.25	82.23	91.39	89.13
U-Net++ [42]	92.19	85.51	93.88	90.56	91.79	84.83	93.85	89.82
FPS-U^{2}Net [12]	90.93	83.37	89.97	91.91	90.79	83.13	92.38	89.25
FDE U-Net [13]	93.57	87.91	94.64	92.52	91.33	84.05	94.07	88.75
DualSight-Fire (Ours)	95.18	90.81	94.83	95.54	92.27	85.65	92.36	92.18

TABLE V

CROSS-DATASET GENERALIZATION RESULTS BETWEEN ACTIVE FIRE AND LAND8FIRE

Training set	Test set	Methods	F1 (%)	IoU (%)	Precision (%)	Recall (%)
Active Fire	Land8Fire	U-Net [7]	90.49	82.63	86.00	95.48
		Attention U-Net [41]	92.31	85.72	90.15	94.58
		U-Net++ [42]	91.67	84.61	87.99	95.66
		FPS-U^{2}Net [12]	83.36	71.46	76.80	91.14
		FDE U-Net [13]	90.87	83.26	90.66	91.07
		DualSight-Fire (Ours)	92.65	86.31	92.45	92.85
Land8Fire	Active Fire	U-Net [7]	84.90	73.76	84.87	84.94
		Attention U-Net [41]	84.49	73.15	84.33	84.65
		U-Net++ [42]	82.02	69.52	81.11	82.95
		FPS-U^{2}Net [12]	81.36	68.58	92.94	72.35
		FDE U-Net [13]	79.92	66.55	81.23	78.64
		DualSight-Fire (Ours)	87.50	77.79	85.87	89.20

3) Performance on Different Fire Scales: To further explore the model's detection capability for fire spots of different scales, we merged the two datasets and divided them into a tiny fire group (<100 pixels) and a large fire group (>100 pixels) based on the number of fire pixels for separate evaluation.

As shown in Table IV, DualSight-Fire performs excellently in both the tiny fire group and the large fire group. Especially in the most challenging tiny fire group, our model achieves an $F1$-score of 95.18% and an IoU of 90.81%, significantly outperforming other comparison methods. This result fully demonstrates the immense advantage of the dual-branch architecture and SFL in solving the problem of small object segmentation.

4) Cross-Dataset Generalization Evaluation: To further evaluate model transferability under dataset shift, we conducted bidirectional cross-dataset experiments between Active Fire and Land8Fire. In each setting, models were trained on one source dataset and tested on the other target dataset. For fair comparison, the operating threshold of each model was selected on the validation split of the target dataset according to the maximum $F1$-score, and the selected threshold was then fixed for evaluation on the target test split.

As shown in Table V, DualSight-Fire achieves the best $F1$-score and IoU in both transfer directions. When trained on Active Fire and tested on Land8Fire, it improves $F1$-score and IoU over U-Net by 2.16% and 3.68%, respectively. When trained on Land8Fire and tested on Active Fire, it improves $F1$-score and IoU over U-Net by 2.60% and 4.03%, respectively. These results indicate that the proposed dual-decoder background modeling mechanism improves cross-dataset robustness under distribution shifts between independently annotated fire datasets.

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TABLE VI

ABLATION STUDY ON THE CONTRIBUTION OF DIFFERENT COMPONENTS. BCE AND FDFM DENOTE BINARY CROSS-ENTROPY AND FEATURE DISCREPANCY FUSION MODULE, RESPECTIVELY

Baseline	BCE	Focal loss	Spatial Focal loss	Reconstruction Decoder	FDFM	F1 (%)	IoU (%)	Precision (%)	Recall (%)
✓	✓					93.38	87.58	94.37	92.41
✓	✓			✓		95.35	91.11	95.46	95.24
✓	✓			✓	✓	95.76	91.86	95.34	96.19
✓		✓				93.88	88.46	97.17	90.80
✓		✓		✓		94.98	90.44	95.47	94.49
✓		✓		✓	✓	95.66	91.69	94.68	95.66
✓			✓			94.30	89.21	97.44	91.35
✓			✓	✓		95.22	90.88	95.29	95.16
✓			✓	✓	✓	95.92	92.16	95.82	96.02

5) Statistical Significance Analysis: To strictly validate the reliability of the performance improvement, we conducted a paired statistical significance analysis using the seed-wise results from the same data split. Because each baseline and DualSight-Fire were evaluated with the same ten random-seed indices, we used the two-sided Wilcoxon signed-rank test to compare DualSight-Fire with each baseline on $F1$-score and IoU. On the Active Fire dataset, the improvements of DualSight-Fire over every baseline are statistically significant for both $F1$-score and IoU, with all paired comparisons yielding $p = 0.002$. On the Land8Fire dataset, the corresponding paired comparisons are also statistically significant, with p-values ranging from 0.002 to 0.010. These results indicate that the observed gains are consistent across repeated training runs and are not attributable to a single favorable random seed.

6) Ablation Study: To evaluate the effectiveness of the key components in DualSight-Fire, we conduct an ablation study on the Active Fire dataset [9], which contains a substantial number of samples. The experimental results are summarized in Table VI. The qualitative results are presented in the Appendix.

a) Effectiveness of dual-decoder architecture: To evaluate the contribution of the reconstruction decoder, we compare the single-stream baselines with their dual-branch counterparts across different loss functions. As shown in Table VI, regardless of the loss function used, the introduction of the reconstruction branch consistently improves performance. For instance, under the SFL setting, adding the reconstruction decoder without FDFM increases the recall from 91.35% to 95.16% and the $F1$-score from 94.30% to 95.22%. Similar improvements are observed under BCE and focal loss, where the $F1$-score increases from 93.38% to 95.35% and from 93.88% to 94.98%, respectively. This effectively validates that the auxiliary background discrimination task forces the shared encoder to learn more discriminative representations, thereby recovering more fire pixels that were previously missed.

b) Effectiveness of feature discrepancy fusion module (FDFM): We further scrutinize the fusion strategy by comparing the simple convolutional fusion (w/o FDFM) with our proposed difference-based fusion (Full Model). The benefit of FDFM is evident across all settings. Under the SFL setting, replacing simple convolutional fusion with FDFM improves the $F1$-score from 95.22% to 95.92%. Consistent gains are also observed under BCE and focal loss, where the $F1$-score increases from 95.35% to 95.76% and from 94.98% to 95.66%, respectively. This validates that the logic of "subtracting background" is more effective than simple concatenation, as it utilizes the background logit map as a mask to filter out semantic false alarms.

c) Effectiveness of SFL: To isolate the contribution of the proposed SFL, we first compare the single-stream baseline trained with BCE and the same baseline trained with SFL, where the network architecture remains unchanged, and only the loss function is replaced. With SFL, the $F1$-score improves from 93.38% to 94.30%, the IoU improves from 87.58% to 89.21%, and the precision increases from 94.37% to 97.44%. This demonstrates that the area-aware and shape-aware weighting terms can improve small-fire and boundary-sensitive learning even without the reconstruction branch or FDFM. When incorporated into the full dual-branch architecture with FDFM, SFL further leads to the best overall $F1$-score and IoU, improving over the BCE-based full counterpart by 0.16% and 0.30%, respectively.

d) Detailed analysis of fusion strategy: To explore the optimal position for feature fusion, we designed a variant model. As shown in Fig. 7, based on the original model, we integrated the feature discrepancy fusion module (FDFM) after every upsampling stage of the decoder, meaning the fused output continues to serve as input for the next stage of the segmentation decoder, forming a multilevel fusion structure.

The test results of this variant model on the Active Fire dataset [9] are: precision 95.79%, recall 95.43%, $F1$ 95.61%, and IoU 91.59%. Although this result is still better than most comparison models, it is slightly lower than the proposed DualSight-Fire (which fuses only at the last layer). This is likely due to the fact that early introduction of background differences in deep features might interfere with the transmission of high-level semantic information to some extent. Therefore, we finally chose the structure with single fusion at the end of the decoder.

7) Sensitivity Analysis: We conducted a series of sensitivity analyses on the hyperparameters involved in the method.

Impact of shape-aware focusing parameter:

To further investigate the impact of the shape-aware mechanism within the proposed SFL, we conducted a sensitivity analysis on the focusing hyperparameter γ_0. This parameter

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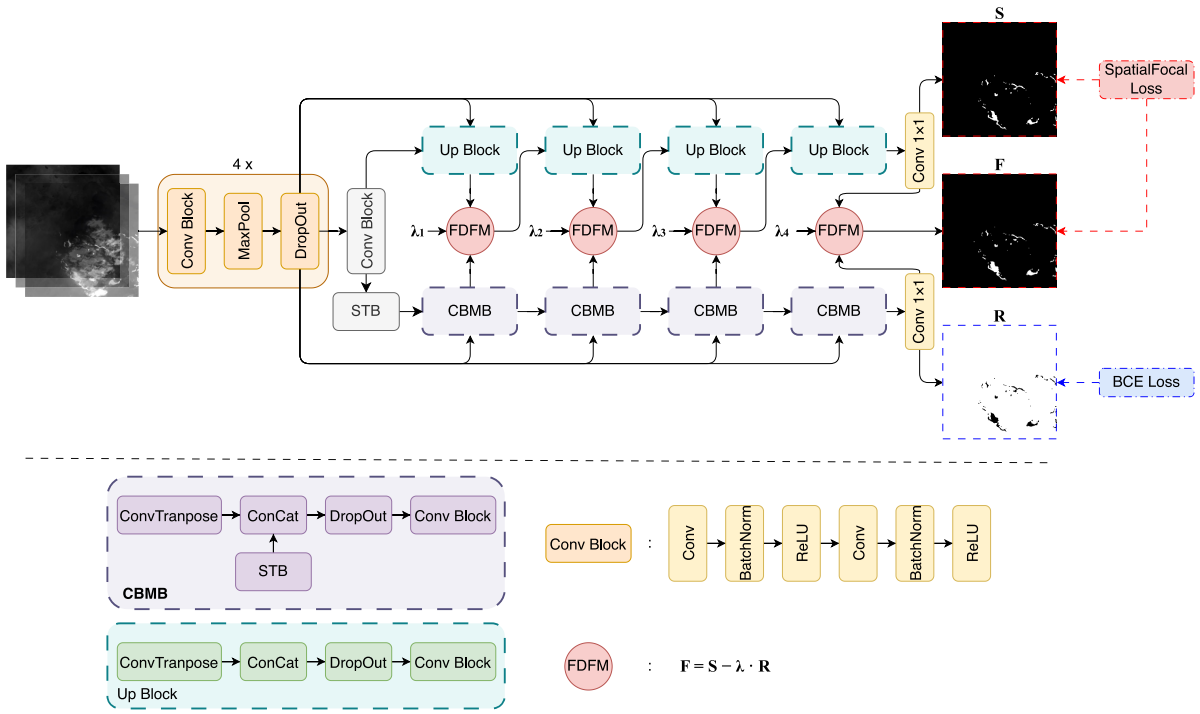


Fig. 7. Structure of the variant model with multilevel fusion.

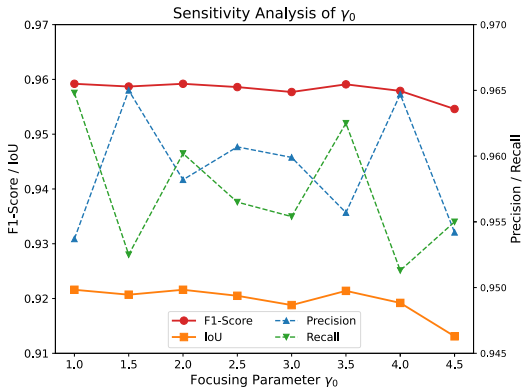


Fig. 8. Sensitivity analysis of the hyperparameter γ_0 in SFL.

TABLE VII
SENSITIVITY ANALYSIS OF THE FOCUSING PARAMETER γ_0

γ_0	F1 (%)	IoU (%)	Precision (%)	Recall (%)
1.0	95.92	92.16	95.37	96.48
1.5	95.87	92.07	96.50	95.25
2.0	95.92	92.16	95.82	96.02
2.5	95.86	92.05	96.07	95.65
3.0	95.77	91.88	95.99	95.54
3.5	95.91	92.14	95.57	96.25
4.0	95.79	91.92	96.47	95.13
4.5	95.46	91.31	95.42	95.50

TABLE VIII
SENSITIVITY ANALYSIS OF THE MULTISCALE LOSS WEIGHTS (λ_{fused}:λ_{seg}:λ_{rec})

Ratio ($\lambda_{fused} : \lambda_{seg} : \lambda_{rec}$)	F1 (%)	IoU (%)	Precision (%)	Recall (%)
0.5 : 0.25 : 0.25	95.77	91.88	95.52	96.02
0.6 : 0.20 : 0.20	95.92	92.16	95.82	96.02
0.7 : 0.15 : 0.15	95.78	91.90	95.67	95.90
0.8 : 0.10 : 0.10	95.84	92.01	95.49	96.19

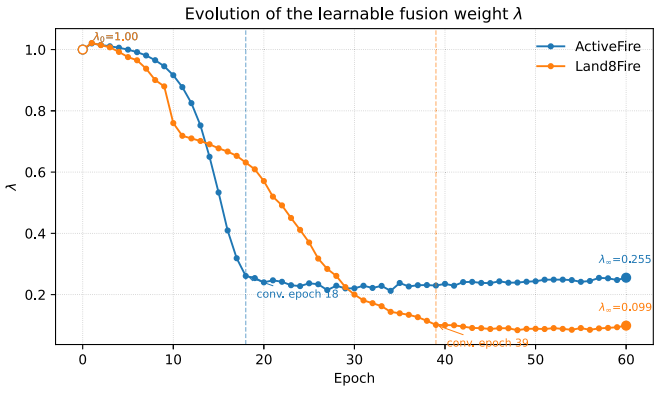


Fig. 9. Evolution of the learnable parameter λ in FDFM during training on the active fire and Land8Fire datasets. Both runs start from $\lambda = 1.0$, undergo a brief rise during the warm-up phase, and then decay to dataset-specific balances.

controls the intensity of the loss function's focus on hard examples, particularly pixels near fire boundaries. We varied γ_0 from 1.0 to 4.5 with a step size of 0.5, while keeping other hyperparameters fixed. The quantitative results are illustrated in Fig. 8 and detailed in Table VII.

As observed from the results, the model exhibits high robustness to variations in γ_0 within the range of $[1.0, 2.5]$. Specifically, when γ_0 is set to 1.0, the model achieves a high recall of 96.48%, indicating strong sensitivity to fire targets, but a slightly lower precision of 95.37%. Increasing

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TABLE IX
PARAMETER SETTINGS

Methods	Bands	Epochs	Batch Size	LR	Loss Function	Optimizer
U-Net [7], U-Net++ [42], Attention U-Net [41]	7, 6, 2	50	16	1e-3	BCE	Adam
FPS-U^{2}Net [12]	7, 6, 6	80	4	1e-3	BCE + IoU (Multi-scale weights: 1 : 1/2 : 1/4 : 1/8 : 1/16 : 1/32)	Adam
FDE U-Net [13]	1–10	100	32	1e-3	BCE	Adam
DualSight-Fire (Ours)	7, 6, 5	60	16	1e-3	Spatial-focal (Seg/Fused) + BCE (Rec) (Multi-scale Weights: Fused:Seg:Rec = 0.6 : 0.2 : 0.2)	Adam

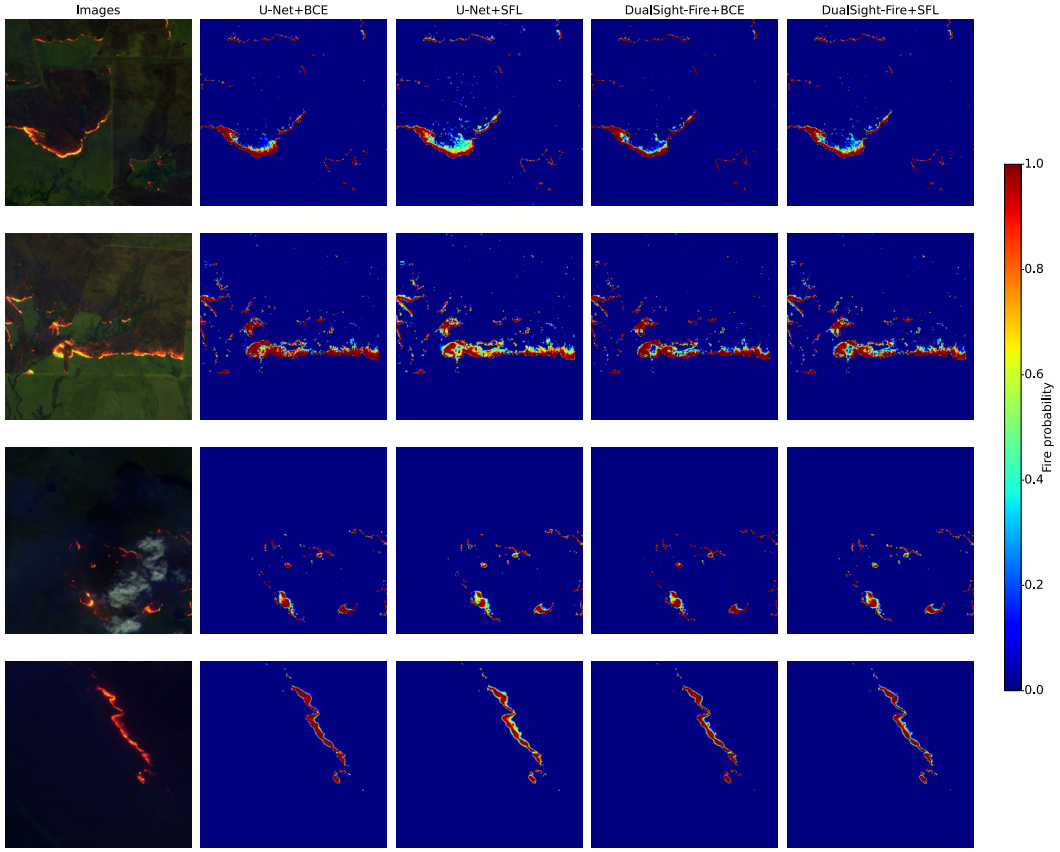


Fig. 10. Comparative analysis of fire probability heatmaps. BCE denotes BCE loss; SFL denotes spatial-focal loss.

γ_0 to 2.0 yields the optimal balance, achieving the highest $F1$-score (95.92%) and IoU (92.16%), with precision improving to 95.82%. This suggests that a moderate focusing strength effectively enhances boundary delineation without suppressing valid fire pixels.

However, a performance degradation is observed when γ_0 exceeds 3.0. For instance, at $\gamma_0 = 4.5$, the $F1$-score drops to 95.46% and IoU to 91.31%. This phenomenon can be attributed to the nature of the focusing mechanism: an excessively large γ_0 may cause the loss function to over-penalize hard examples and ambiguous boundary pixels, leading to optimization instability or overfitting to labeling noise. Consequently, we empirically selected $\gamma_0 = 2.0$ as the default setting for all experiments to ensure a robust tradeoff between precision and recall.

Impact of multi-task loss weighting:

In the proposed joint optimization objective (5), the balancing weights λ_{fused}, λ_{seg}, and λ_{rec} determine the relative importance of the final fusion output versus the auxiliary supervision from individual branches. To identify the optimal tradeoff, we conducted experiments with different weight ratios, keeping the segmentation and reconstruction weights equal ($\lambda_{seg} = \lambda_{rec}$) while varying the weight of the fused output λ_{fused}.

The results are presented in Table VIII. It can be observed that the model achieves the best overall performance (95.92% $F1$ and 92.16% IoU) when the ratio is set to 0.6:0.2:0.2. When λ_{fused} is relatively low (e.g., 0.5), the optimization focus is dispersed, preventing the fusion module from being fully trained to correct the errors of the subbranches. Conversely, when λ_{fused} is excessively high (e.g., 0.8), the supervision signal for the auxiliary branches becomes too weak ($\lambda_{rec} = 0.1$). This weakens the explicit background modeling capability of the reconstruction decoder, leading to a decrease in precision (from 95.82% to 95.49%) and a slight drop in IoU, as the model loses the regularization benefit of the auxiliary tasks.

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TABLE X
SENSITIVITY ANALYSIS OF THE BACKGROUND WEIGHT α_{bg} ON THE ACTIVE FIRE VALIDATION SET

α_{bg}	F1 (%)	IoU (%)	Precision (%)	Recall (%)
0.10	94.78	90.08	91.44	98.37
0.25	95.58	91.54	94.60	96.59
0.50	95.92	92.16	95.82	96.02
0.75	95.37	91.16	96.81	93.98
1.00	93.85	88.42	97.68	90.32

Therefore, a ratio of 0.6:0.2:0.2 provides the optimal balance between refining the final result and maintaining robust feature learning in the subbranches.

V. CONCLUSION

In this article, we propose DualSight-Fire, a cognitively inspired dual-decoder network for robust small fire segmentation in remote sensing imagery. By mimicking the human visual process of foreground spotting and background confirmation, DualSight-Fire integrates a segmentation branch with a background reconstruction branch. The proposed CBMB effectively captures long-range semantic context, while the feature discrepancy fusion module leverages the discrepancy between fire and background predictions to suppress confusing nonfire thermal anomalies. Furthermore, we propose SFL, which dynamically adjusts learning weights based on the area and shape of fire instances, significantly improving the sensitivity to tiny fire spots and boundaries. Extensive experiments on the Active Fire and Land8Fire datasets demonstrated that DualSight-Fire achieves state-of-the-art mean performance across repeated random-seed runs, and paired Wilcoxon signed-rank tests further support the statistical robustness of the improvements in $F1$-score and IoU.

The present study focuses on static fire segmentation from individual multispectral satellite frames, and does not claim to model the continuous temporal evolution of active fires. Temporal changes between Landsat-8 revisits may affect practical fire monitoring if the method is used alone for rapidly spreading fires. Once continuous-time or multisource imagery of the same fire scene is available, DualSight-Fire can be applied frame by frame to construct fire evolution sequences, or serve as the high-spatial-resolution component in a fusion system that combines high-temporal-resolution but lower spatial-resolution sensors. Extending the framework toward dynamic fire monitoring and cross-sensor temporal fusion is therefore an important direction for future work.

APPENDIX

A. Evaluation Metrics

Precision measures the accuracy of positive predictions, reflecting the model's capability to suppress false alarms, while recall assesses the proportion of actual fire pixels correctly identified, indicating the sensitivity to fire targets. The $F1$-score, defined as the harmonic mean of precision and recall, offers a balanced view of the model's overall effectiveness. Additionally, IoU quantifies the structural overlap between the predicted segmentation mask and the ground truth, serving as a rigorous measure of shape accuracy. The mathematical formulations for these metrics are given by

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}}, \quad \text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{6}$$

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}, \quad \text{IoU} = \frac{\text{TP}}{\text{TP}+\text{FP}+\text{FN}} \tag{7}$$

where TP, FP, and FN denote the counts of true-positive, false-positive, and false-negative pixels, respectively.

B. Hyperparameter Setting

To ensure fair comparison, the training strategies and parameter settings for all comparison models strictly followed the recommended configurations in the corresponding original publications. The detailed hyperparameter settings for our DualSight-Fire and other models are shown in Table IX.

C. Convergence Analysis of the Learnable Parameter λ

The feature-difference fusion module (FDFM) introduced in Section III defines the fused representation as $\mathbf{F} = \mathbf{S} - \lambda \cdot \mathbf{R}$, where λ is a learnable parameter that controls how strongly the reconstructed background response $\mathbf{R}$ is subtracted from the segmentation response $\mathbf{S}$. During training, λ is driven jointly by the SFL applied to the fused output, which is weighted by $\lambda_{\text{fused}} = 0.6$ in the total loss, and by the BCE supervision that regresses $\mathbf{R}$ toward the inverse mask $1 - \mathbf{Y}$.

We record λ at the end of every epoch over the 60 training epochs on both datasets, and plot the resulting trajectories in Fig. 9. On Active Fire, λ starts from the initial value 1.0 and briefly rises to 1.020 at the first epoch; it first stays on a short plateau near 1.0 over epochs 0–4, then enters a rapid decay phase over epochs 5–18, and finally settles into a small-amplitude oscillation phase over epochs 19–59. The minimum value 0.2123 is attained at epoch 34, the value at the final epoch (epoch 59) is 0.2553, and the mean over the final ten epochs is 0.249. On Land8Fire, λ similarly starts from 1.0 and briefly rises to 1.022 at the first epoch; it first remains on a plateau over epochs 0–4, then enters a progressive decay phase over epochs 5–30, and finally settles into a low-level stable phase over epochs 31–59. The minimum value 0.0839 is attained at epoch 48, the value at the final epoch (epoch 59) is 0.0988, and the mean over the final ten epochs is 0.090.

From the gradient perspective, the chain rule yields $\partial \mathcal{L}_{total}/\partial \lambda = \sum_{i,j} (\partial \mathcal{L}_{fused}/\partial \mathbf{F}_{i,j}) \cdot (-\mathbf{R}_{i,j})$. Because $\mathbf{R}$ is explicitly supervised by BCE toward $1 - \mathbf{Y}$, λ is updated under a signed weighting that couples the fused-output gradient with the learned background prior, and it converges to the value at which the Spatial-focal cost on $\mathbf{F}$ reaches a local minimum.

These trajectories show that making λ learnable removes the need to tune the background-subtraction strength by hand. On both datasets, λ leaves its initial value 1.0 and settles at a strictly positive balance inside the open interval $(0, 1)$, neither collapsing to 0, which would disable background

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subtraction, nor remaining at 1, which would correspond to equally weighted subtraction. The learned balance values are clearly different between Active Fire and Land8Fire, which indicates that the FDFM adapts the background-subtraction strength to the statistics of each dataset under the same network configuration.

D. Sensitivity Analysis of the Background Weight α_{Bg}

The background weight α_{bg} in the SFL controls the base penalty assigned to background pixels, and therefore directly shapes the tradeoff between precision and recall under the severe foreground–background imbalance of small-fire segmentation. We use $\alpha_{bg} = 0.5$ throughout the experiments reported above, and this section reports a sensitivity analysis on the Active Fire validation set that supports this value. We vary α_{bg} over $\{0.10, 0.25, 0.50, 0.75, 1.00\}$ while keeping all other hyperparameters identical to Table IX, and train each variant for 60 epochs.

The results reported in Table X show clear trends for precision and recall. As α_{bg} increases from 0.10 to 1.00, precision steadily rises from 91.44% to 97.68%, while recall steadily falls from 98.37% to 90.32%; at $\alpha_{bg} = 0.50$, precision and recall are 95.82% and 96.02%, respectively, where the two metrics differ by only 0.20 percentage points. $F1$-score and IoU attain their maximum at $\alpha_{bg} = 0.50$, reaching 95.92% and 92.16%, respectively, and decrease on both sides of this value. A smaller α_{bg} lowers the contribution of background negatives to the total loss, so the network retrieves more fire pixels at the cost of more FPs. A larger α_{bg} raises that contribution, suppresses the gradient signal from tiny fire connected components, and trades recall for precision. $\alpha_{bg} = 0.50$ thus jointly maximizes $F1$-score and IoU on this grid, which supports the value used in our main experiments.

E. Qualitative Results of the Ablation Study

To visually demonstrate the contribution of each component to model performance, we provide a comparative analysis of fire probability heatmaps under different configurations in Fig. 10. Compared to the baseline U-Net model, DualSight-Fire with the dual-decoder architecture exhibits significant feature enhancement and noise suppression capabilities. The heatmaps show that the dual-decoder structure significantly increases the predicted probability of high-confidence fire regions (fire pixels turn from yellow to red) while effectively suppressing noise response in low-confidence background regions (background pixels turn from green to blue). This confirms that the reconstruction branch effectively enhances the model's foreground–background discrimination ability in complex scenes through explicit background modeling. Compared to traditional BCE Loss, SFL excels in handling small objects and boundary details thanks to its unique spatial topology perception mechanism (i.e., area awareness and boundary awareness). The heatmaps show that after applying SFL, tiny connected fire spots that were originally missed (green pixels) are successfully recalled, and the boundary contours of large fire areas are more clearly defined. This indicates that the loss function successfully guides the model to focus on difficult-to-classify tiny targets and boundary pixels through dynamic weighting. In summary, the organic combination of the DualSight-Fire architecture and SFL not only strengthens the model's confidence in identifying significant fire spots but also overcomes the challenges of missed detection of tiny targets and blurred boundary segmentation, achieving precise pixel-wise classification of fire targets.

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